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Practical 1

Aim : Install, configure and run Hadoop and HDFS and explore HDFS.

Steps to Install Hadoop

1. Install Java JDK 1.8
2. Download Hadoop and extract and place under C drive
3. Set Path in Environment Variables
4. Config files under Hadoop directory
5. Create folder datanode and namenode under data directory
6. Edit HDFS and YARN files
7. Set Java Home environment in Hadoop environment
8. Setup Complete. Test by executing start-all.cmd

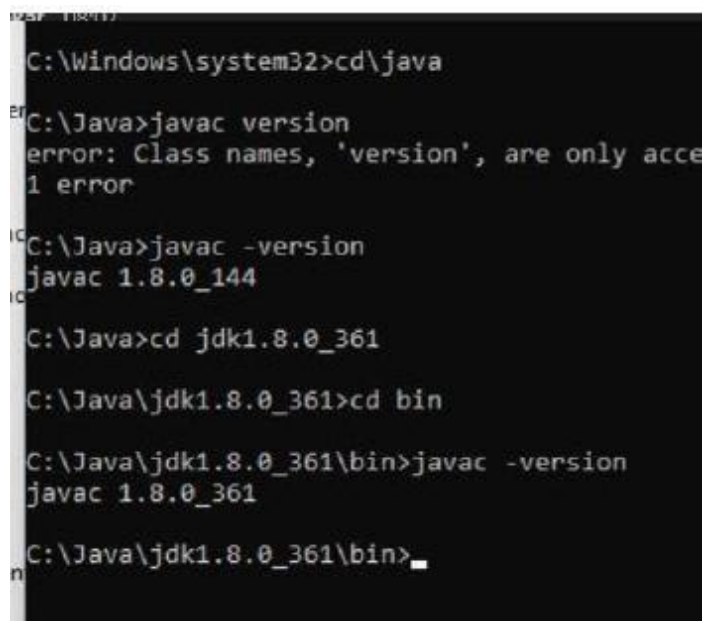
There are two ways to install Hadoop, i.e.

9. Single node
10. Multi node

Here, we use multi node cluster.

1. Install Java

- – Java JDK Link to download
 - o <https://www.oracle.com/java/technologies/javase-jdk8-downloads.html>
- – extract and install Java in C:\Java
- – open cmd and type -> javac -version

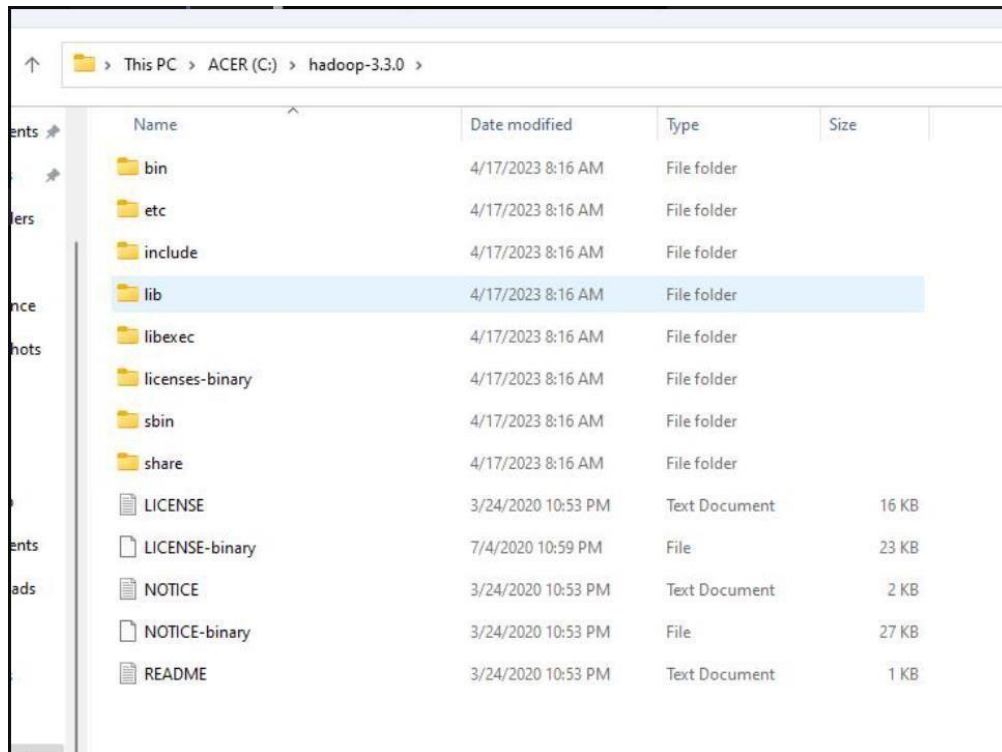


```
C:\Windows\system32>cd\java
C:\Java>javac version
error: Class names, 'version', are only accepted if used in a batch file
1 error
C:\Java>javac -version
javac 1.8.0_144
C:\Java>cd jdk1.8.0_361
C:\Java\jdk1.8.0_361>cd bin
C:\Java\jdk1.8.0_361\bin>javac -version
javac 1.8.0_361
C:\Java\jdk1.8.0_361\bin>
```

2. Download Hadoop

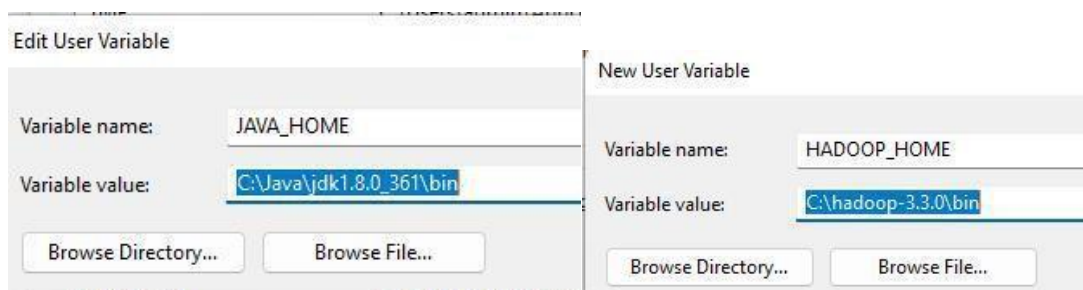
<https://www.apache.org/dyn/closer.cgi/hadoop/common/hadoop-3.3.0/hadoop-3.3.0.tar.gz>

- right click .rar.gz file -> show more options -> 7-zip->and extract to C:\Hadoop-3.3.0\



3. Set the path JAVA_HOME Environment variable

4. Set the path HADOOP_HOME Environment variable Click on New to both user variables and system variables. Click on user variable -> path -> edit-> add path for Hadoop and java upto 'bin'



Click Ok, Ok, Ok.

5. Configurations

Edit file C:/Hadoop-3.3.0/etc/hadoop/core-site.xml,

paste the xml code in folder and save

```
<configuration>
<property>
<name>fs.defaultFS</name>
<value>hdfs://localhost:9000</value>
</property>
</configuration>
```

=====

Rename “mapred-site.xml.template” to “mapred-site.xml” and edit this file C:/Hadoop-3.3.0/etc/hadoop/mapred-site.xml, paste xml code and save this file.

=====

```
<configuration>
<property>
<name>mapreduce.framework.name</name>
<value>yarn</value>
</property>
</configuration>
```

=====

Create folder “data” under “C:\Hadoop-3.3.0”

Create folder “datanode” under “C:\Hadoop-3.3.0\data”

Create folder “namenode” under “C:\Hadoop-3.3.0\data”

=====

Edit file C:\Hadoop-3.3.0/etc/hadoop/hdfs-site.xml,

paste xml code and save this file.

```
<configuration>
<property>
<name>dfs.replication</name>
<value>1</value>
</property>
<property>
```

```
<name>dfs.namenode.name.dir</name>
<value>/hadoop-3.3.0/data/namenode</value>
</property>
<property>
<name>dfs.datanode.data.dir</name>
<value>/hadoop-3.3.0/data/datanode</value>
</property>
</configuration>
```

=====

**Edit file C:/Hadoop-3.3.0/etc/hadoop/yarn-site.xml,
paste xml code and save this file.**

```
<configuration>
<property>
<name>yarn.nodemanager.aux-services</name>
<value>mapreduce_shuffle</value>
</property>
<property>
<name>yarn.nodemanager.auxservices.mapreduce.shuffle.class</name>
<value>org.apache.hadoop.mapred.ShuffleHandler</value>
</property>
<property>
<name>yarn.resourcemanager.address</name>
<value>127.0.0.1:8032</value>
</property>
<property>
<name>yarn.resourcemanager.scheduler.address</name>
<value>127.0.0.1:8030</value>
</property>
<property>
<name>yarn.resourcemanager.resource-tracker.address</name>
<value>127.0.0.1:8031</value>
```

```
</property>
```

```
</configuration>
```

```
=====
```

6. Edit file C:/Hadoop-3.3.0/etc/hadoop/hadoop-env.cmd

Find "JAVA_HOME=%JAVA_HOME%" and replace it as

```
set JAVA_HOME="C:\Java\jdk1.8.0_361"
```

```
=====
```

7. Download "redistributable" package

Download and run VC_redist.x64.exe

8. Hadoop Configurations

Download bin folder from <https://github.com/s911415/apache-hadoop-3.1.0-winutils>

– Copy the bin folder to c:\hadoop-3.3.0. Replace the existing bin folder.

9. copy "hadoop-yarn-server-timelineservice-3.0.3.jar" from ~\hadoop-3.0.3\share\hadoop\yarn\timelineservice to ~\hadoop-3.0.3\share\hadoop\yarn folder.

10. Format the NameNode

– Open cmd 'Run as Administrator' and type command "hdfs namenode –format"

```

2023-04-17 09:14:34,324 INFO util.GSet: VM type = 64-bit
2023-04-17 09:14:34,324 INFO util.GSet: 0.029999999329447746% max memory 889 MB = 273.1 KB
2023-04-17 09:14:34,325 INFO util.GSet: capacity = 2^15 = 32768 entries
Re-format filesystem in Storage Directory root= C:\hadoop-3.3.0\data\namenode; location= null ? (Y or N) y
2023-04-17 09:14:55,710 INFO namenode.FSImage: Allocated new BlockPoolId: BP-813810208-169.254.162.181-16817
2023-04-17 09:14:55,711 INFO common.Storage: Will remove files: [C:\hadoop-3.3.0\data\namenode\current\fsimage_00000000000000000000000000000000.md5, C:\hadoop-3.3.0\data\name
\seen_txid, C:\hadoop-3.3.0\data\namenode\current\VERSION]
2023-04-17 09:14:55,752 INFO common.Storage: Storage directory C:\hadoop-3.3.0\data\namenode has been succes
tted.
2023-04-17 09:14:55,775 INFO namenode.FSImageFormatProtobuf: Saving image file C:\hadoop-3.3.0\data\namenode
image.ckpt_00000000000000000000 using no compression
2023-04-17 09:14:55,840 INFO namenode.FSImageFormatProtobuf: Image file C:\hadoop-3.3.0\data\namenode\current
pt_00000000000000000000 of size 397 bytes saved in 0 seconds .
2023-04-17 09:14:55,856 INFO namenode.NNStorageRetentionManager: Going to retain 1 images with txid >= 0
2023-04-17 09:14:55,862 INFO namenode.FSImage: FSImageSaver clean checkpoint: txid=0 when meet shutdown.
2023-04-17 09:14:55,863 INFO namenode.NameNode: SHUTDOWN_MSG:
/*****
SHUTDOWN_MSG: Shutting down NameNode at VSIT-X121-15/169.254.162.181
*****/
C:\hadoop-3.3.0\bin>

```

11. Testing

– Open cmd 'Run as Administrator' and change directory to C:\Hadoop-3.3.0\sbin

– type start-all.cmd

OR

- type start-dfs.cmd

– type start-yarn.cmd

```

C:\hadoop-3.3.0\sbin>start-all.cmd
This script is Deprecated. Instead use start-dfs.cmd and start-yarn.cmd
The filename, directory name, or volume label syntax is incorrect.
The filename, directory name, or volume label syntax is incorrect.
starting yarn daemons
The filename, directory name, or volume label syntax is incorrect.

```

– You will get 4 more running threads for Datanode, namenode, resource manager and node manager

The image shows four terminal windows from the Apache Hadoop Distribution, illustrating the startup process of different components:

- namenode:** Shows logs for the NameNode starting, including RPC server initialization, webapp startup, and registration with the ResourceManager.
- jps:** Displays the current Java processes running: 8164 NameNode, 11272 Jps, 11576 NodeManager, and 2616 DataNode.
- resourcemanager:** Shows logs for the ResourceManager starting, including IPC server initialization, resource tracker service registration, and transition to an active state.
- datanode:** Shows logs for the DataNode starting, including directory scanner initialization, block pool creation, and successful registration with the NameNode.

Output:

12. Type JPS command to start-all.cmd command prompt, you will get following output.

```

C:\hadoop-3.3.0\sbin>jps
8164 NameNode
11272 Jps
11576 NodeManager
2616 DataNode
2952 ResourceManager

C:\hadoop-3.3.0\sbin>

```

13. Run <http://localhost:9870/> from any browser

The screenshot shows the Hadoop Overview page in a web browser. The page has a green navigation bar with tabs: Hadoop, Overview (selected), Datanodes, Datanode Volume Failures, Snapshot, Startup Progress, and Utilities. The main content area is titled "Overview 'localhost:9000' (✓active)". Below the title is a table with the following information:

Started:	Mon Apr 17 09:18:18 +0530 2023
Version:	3.3.0, raa96f1871bfd858f9bac59cf2a81ec470da649af
Compiled:	Tue Jul 07 00:14:00 +0530 2020 by brahma from branch-3.3.0
Cluster ID:	CID-7483febe-8254-46e7-bac1-9e1d4d8cbee0
Block Pool ID:	BP-813810208-169.254.162.181-1681703095701

Below the table is a section titled "Summary" with the text "Security is off." at the bottom. The browser's taskbar at the bottom shows the Windows logo, search icon, and various application icons. The system tray on the right shows "ENG IN", a network icon, and the date/time "9:21 AM 4/17/2023".

This is a duplicate of the screenshot above, showing the Hadoop Overview page with the same navigation bar, table of cluster details, and summary information. The browser's taskbar and system tray also match the first screenshot.

Practical 2

Aim: Implement word count / frequency programs using MapReduce

Solution:

C:\hadoop-3.3.0\sbin>start-dfs.cmd

C:\hadoop-3.3.0\sbin>start-yarn.cmd

Open a command prompt as administrator and run the following command to create an input and output folder on the Hadoop file system, to which we will be moving the sample.txt file for our analysis.

C:\hadoop-3.3.0\bin>cd\

C:\>hadoop dfsadmin -safemode leave

DEPRECATED: Use of this script to execute hdfs command is deprecated.

Instead use the hdfs command for it.

Safe mode is OFF

C:\>hadoop fs -mkdir /input_dir

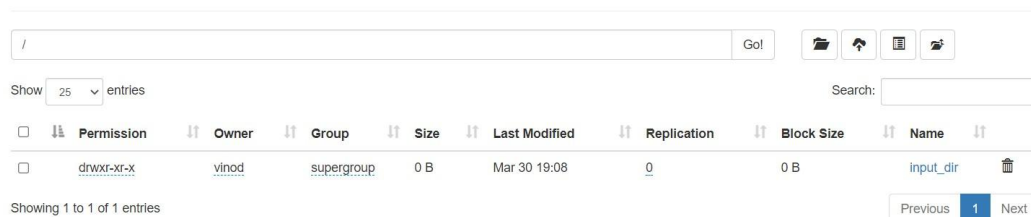
Check it by giving the following URL at browser

<http://localhost:9870>

Utilities -> browse the file system



Browse Directory



Copy the input text file named input_file.txt in the input directory (input_dir) of HDFS.

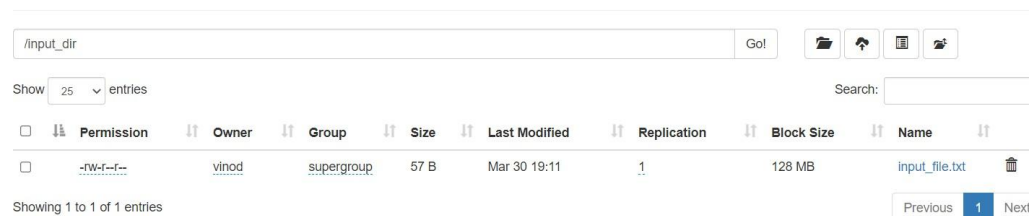
Make a file in c:\input_file.txt and write following content in it.

Hadoop Window version is easy compared to Ubuntu version

Now apply the following command at c:\>

C:\>hadoop fs -put C:/input_file.txt /input_dir

Browse Directory



Verify input_file.txt available in HDFS input directory (input_dir).

C:\>Hadoop fs -ls /input_dir/

```
C:\>hadoop fs -put C:/input_file.txt /input_dir

C:\>hadoop fs -ls /input_dir/
Found 1 items
-rw-r--r--  1 vinod supergroup          57 2023-03-30 19:11 /input_dir/input_file.txt

C:\>
```

Verify content of the copied file

```
C:\>hadoop dfs -cat /input_dir/input_file.txt
```

You can see the file content displayed on the CMD.

```
C:\>hadoop dfs -cat /input_dir/input_file.txt
DEPRECATED: Use of this script to execute hdfs command is deprecated.
Instead use the hdfs command for it.
Hadoop Window version is easy compared to Ubuntu version.
C:\>
```

Run MapReduceClient.jar and also provide input and out directories.

```
C:\>hadoop jar C:/hadoop-3.3.0/share/hadoop/mapreduce/hadoop-mapreduce-examples-3.3.0.jar wordcount /input_dir /output_dir
```

```
Reduce input groups=8
Reduce shuffle bytes=103
Reduce input records=8
Reduce output records=8
Spilled Records=16
Shuffled Maps =1
Failed Shuffles=0
Merged Map outputs=1
GC time elapsed (ms)=70
CPU time spent (ms)=219
Physical memory (bytes) snapshot=517128192
Virtual memory (bytes) snapshot=792633344
Total committed heap usage (bytes)=392691712
Peak Map Physical memory (bytes)=314761216
Peak Map Virtual memory (bytes)=465485824
Peak Reduce Physical memory (bytes)=202366976
Peak Reduce Virtual memory (bytes)=327180288

Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0

File Input Format Counters
  Bytes Read=56

File Output Format Counters
  Bytes Written=65

C:\Windows\System32>
```

In case, there is some error in executing then copy the file MapReduceClient.jar in C:\ and run the program with the jar file using existing MapReduceClient.jar file as:

```
C:\>hadoop jar C:/MapReduceClient.jar wordcount /input_dir /output_dir
```

Now, check the output_dir on browser as follows:



Browse Directory

/

Show 25 entries Search:

☐	Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name	☐
☐	drwxr-xr-x	vinod	supergroup	0 B	Mar 30 19:29	0	0 B	input_dir	☐
☐	drwxr-xr-x	vinod	supergroup	0 B	Mar 30 19:30	0	0 B	output_dir	☐

Click on output_dir → part-r-00000 → Head the file (first 32 K) and check the file content as the output.

File information - part-r-00000

[Download](#)
[Head the file \(first 32K\)](#)
[Tail the file \(last 32K\)](#)

Block information -- Block 0

Block ID: 1073741832
 Block Pool ID: BP-537931513-192.168.1.19-1680861805234
 Generation Stamp: 1008
 Size: 65
 Availability:
 • DESKTOP-OL8EULH

File contents

```
Hadoop 1
Ubuntu 1
Window 1
compared 1
easy 1
is 1
to 1
version 2
```

Search:

Block Size	Name	
MB	_SUCCESS	☐
MB	part-r-00000	☐

[Previous](#)
[1](#)
[Next](#)

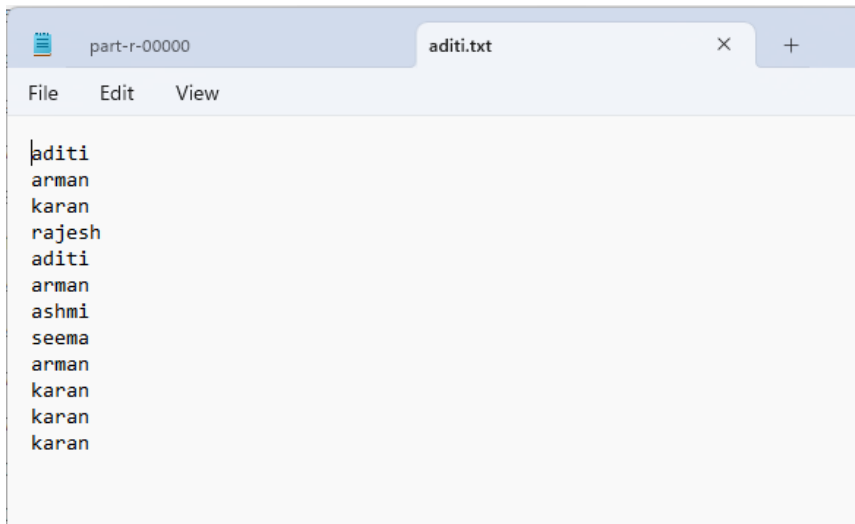
Alternatively, you may type the following command on CMD window as:

```
C:\> hadoop dfs -cat /output_dir/*
```

You can get the following output

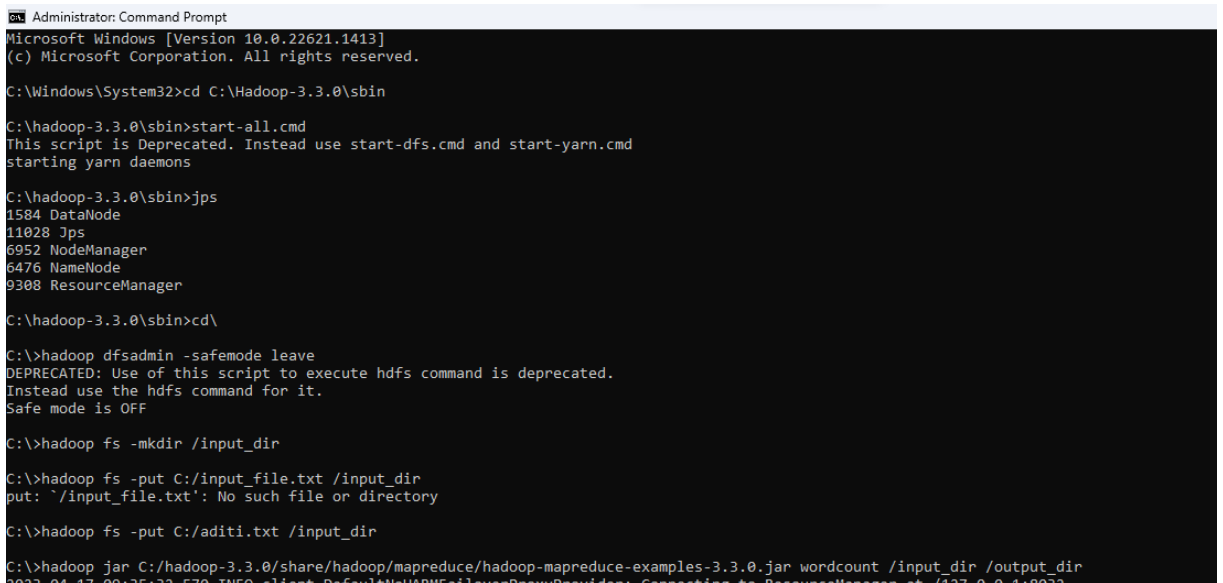
```
C:\Windows\System32>hadoop dfs -cat /output_dir/*
DEPRECATED: Use of this script to execute hdfs command is deprecated.
Instead use the hdfs command for it.
Hadoop 1
Ubuntu 1
Window 1
compared 1
easy 1
is 1
to 1
version 2
C:\Windows\System32>
```

Output:



The screenshot shows a text editor window with a single tab titled 'aditi.txt'. The menu bar includes 'File', 'Edit', and 'View'. The text content of the file is as follows:

```
aditi
arman
karan
rajesh
aditi
arman
ashmi
seema
arman
karan
karan
karan
```



The screenshot shows a Windows Command Prompt window titled 'Administrator: Command Prompt'. The text content is as follows:

```
Microsoft Windows [Version 10.0.22621.1413]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>cd C:\Hadoop-3.3.0\sbin

C:\hadoop-3.3.0\sbin>start-all.cmd
This script is Deprecated. Instead use start-dfs.cmd and start-yarn.cmd
starting yarn daemons

C:\hadoop-3.3.0\sbin>jps
1584 DataNode
11028 Jps
6952 NodeManager
6476 NameNode
9308 ResourceManager

C:\hadoop-3.3.0\sbin>cd\

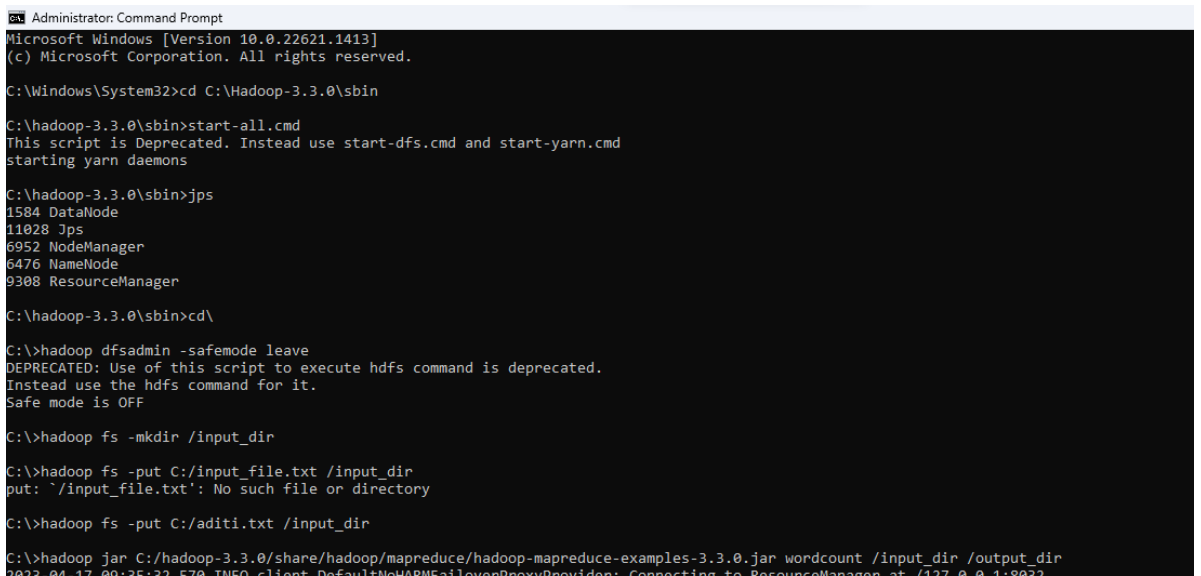
C:\>hadoop dfsadmin -safemode leave
DEPRECATED: Use of this script to execute hdfs command is deprecated.
Instead use the hdfs command for it.
Safe mode is OFF

C:\>hadoop fs -mkdir /input_dir

C:\>hadoop fs -put C:/input_file.txt /input_dir
put: '/input_file.txt': No such file or directory

C:\>hadoop fs -put C:/aditi.txt /input_dir

C:\>hadoop jar C:/hadoop-3.3.0/share/hadoop/mapreduce/hadoop-mapreduce-examples-3.3.0.jar wordcount /input_dir /output_dir
2023-04-17 09:35:32,570 INFO client.DefaultHadoopRMFailoverProxyProvider: Connecting to ResourceManager at /127.0.0.1:8032
```



The screenshot shows a Windows Command Prompt window titled 'Administrator: Command Prompt'. The text content is as follows:

```
Microsoft Windows [Version 10.0.22621.1413]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>cd C:\Hadoop-3.3.0\sbin

C:\hadoop-3.3.0\sbin>start-all.cmd
This script is Deprecated. Instead use start-dfs.cmd and start-yarn.cmd
starting yarn daemons

C:\hadoop-3.3.0\sbin>jps
1584 DataNode
11028 Jps
6952 NodeManager
6476 NameNode
9308 ResourceManager

C:\hadoop-3.3.0\sbin>cd\

C:\>hadoop dfsadmin -safemode leave
DEPRECATED: Use of this script to execute hdfs command is deprecated.
Instead use the hdfs command for it.
Safe mode is OFF

C:\>hadoop fs -mkdir /input_dir

C:\>hadoop fs -put C:/input_file.txt /input_dir
put: '/input_file.txt': No such file or directory

C:\>hadoop fs -put C:/aditi.txt /input_dir

C:\>hadoop jar C:/hadoop-3.3.0/share/hadoop/mapreduce/hadoop-mapreduce-examples-3.3.0.jar wordcount /input_dir /output_dir
2023-04-17 09:35:32,570 INFO client.DefaultHadoopRMFailoverProxyProvider: Connecting to ResourceManager at /127.0.0.1:8032
```

Practical 3

Aim : Implement an application that stores big data in Hbase / MongoDB and manipulate it using R / Python

a. PyMongo

b. Mongo Database

Step A: Install Mongo database

Step 1) Go to (<https://www.mongodb.com/download-center/community>) and Download MongoDB Community Server. We will install the 64-bit version for Windows.

Step 2) Once download is complete open the msi file. Click Next in the start up screen

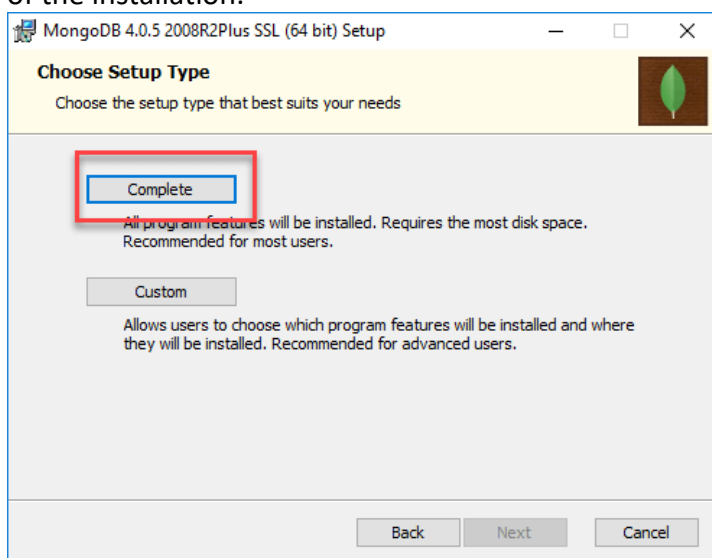


Step 3)

1. Accept the End-User License Agreement

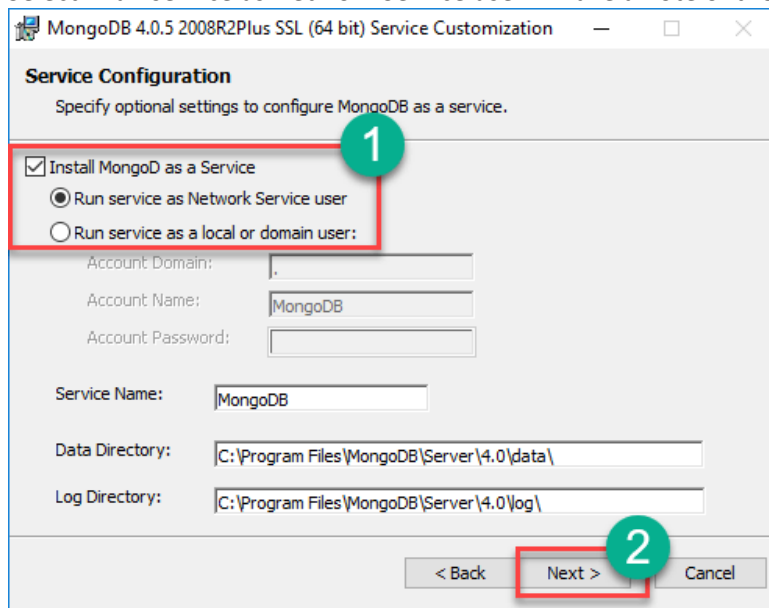
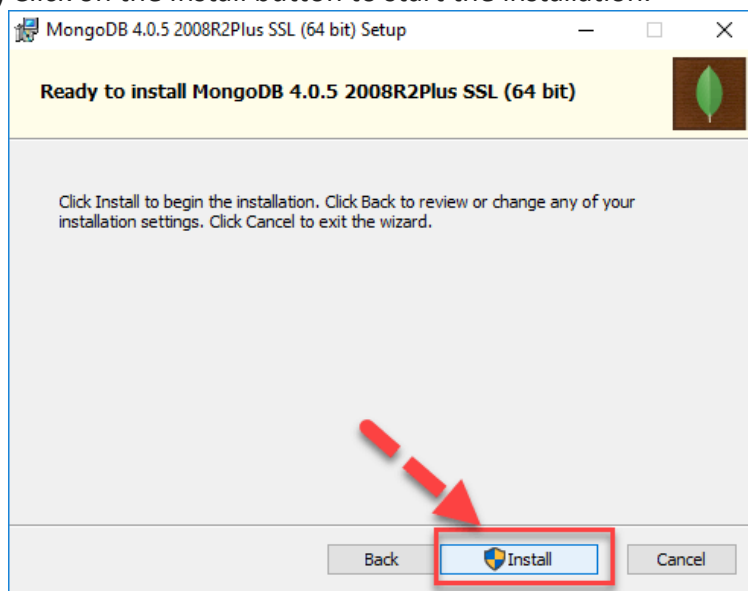
2. Click Next

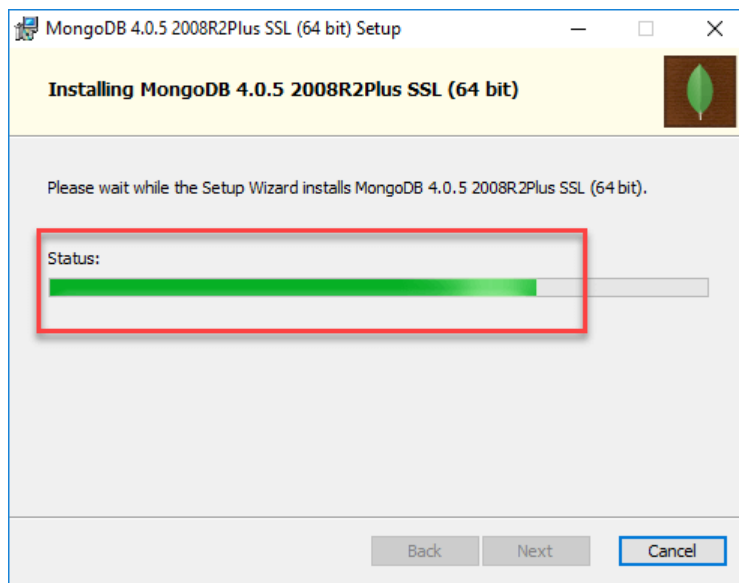
Step 4) Click on the "complete" button to install all of the components. The custom option can be used to install selective components or if you want to change the location of the installation.



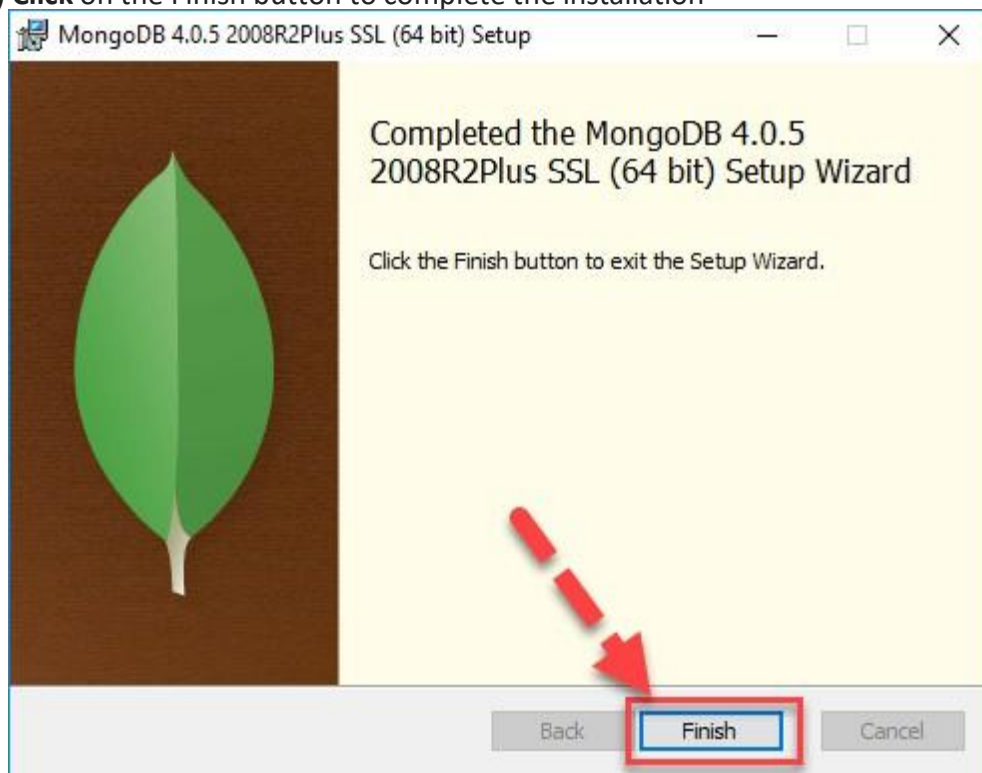
Step 5)

1. Select "Run service as Network Service user". make a note of the data directory,

**Step 6)** Click on the Install button to start the installation.**Step 7)** Installation begins. Click Next once completed.



Step 8) Click on the Finish button to complete the installation



Test Mongoddb

Step 1) Go to " C:\Program Files\MongoDB\Server\4.0\bin" and double click on **mongo.exe**.

Alternatively, you can also click on the MongoDB desktop icon.

Create the directory where MongoDB will store its files.

Open command prompt window and apply following commands

```
C:\users\admin> cd\
```

```
C:\>md data\db
```

```
2023-04-19T08:03:55.694+0530 I INDEX [LogicalSessionCacheRefresh] build index on: config.system.s
{ v: 2, key: { lastUse: 1 }, name: "lsidTTLIndex", ns: "config.system.sessions", expireAfterSeconds:
2023-04-19T08:03:55.695+0530 I INDEX [LogicalSessionCacheRefresh] building index using bulk m
porarily use up to 500 megabytes of RAM
2023-04-19T08:03:55.699+0530 I INDEX [LogicalSessionCacheRefresh] build index done. scanned 0 to
```


Step 2) Execute mongod

Open another command prompt window.

```
C:\> cd C:\Program Files\MongoDB\Server\4.0\bin
```

```
C:\Program Files\MongoDB\Server\4.0\bin> mongod
```

In case if it gives an error then run the following command:

```
C:\Program Files\MongoDB\Server\4.0\bin> mongod -repair
```

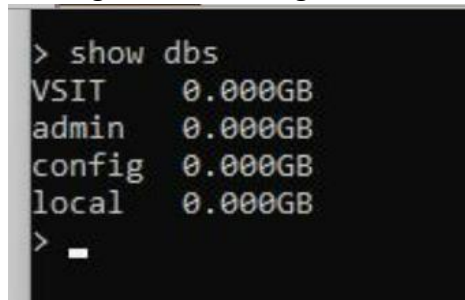
Step 3) Connect to MongoDB using the Mongo shell

Let the MongoDB daemon to run.

Open another command prompt window and run the following commands:

```
C:\users\admin> cd C:\Program Files\MongoDB\Server\4.0\bin
```

```
C:\Program Files\MongoDB\Server\4.0\bin> mongo
```



```
> show dbs
VSIT      0.000GB
admin     0.000GB
config    0.000GB
local     0.000GB
> _
```

Step 4) Install PyMongo

Open another command prompt window and run the following commands:

Check the python version on your desktop / laptop and copy that path from window explorer

```
C:\users\admin>cd C:\Program Files\Python311\Scripts
```

```
C:\Program Files\Python38\Scripts > python -m pip install pymongo
```

Step 5) Test PyMongo

Run the following command from python command prompt

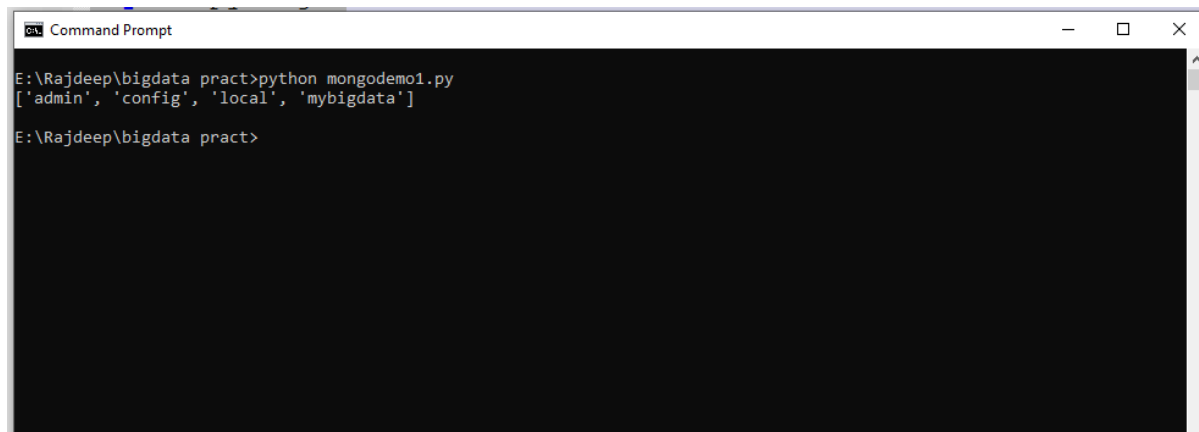
```
import pymongo
```

Now, either create a file in Python IDLE or run all commands one by one in sequence on Python cell

Program 1: Displaying the database name:

```
import pymongo
myclient = pymongo.MongoClient("mongodb://localhost:27017/")
mydb = myclient["mybigdata"]
print(myclient.list_database_names())
```

Output:

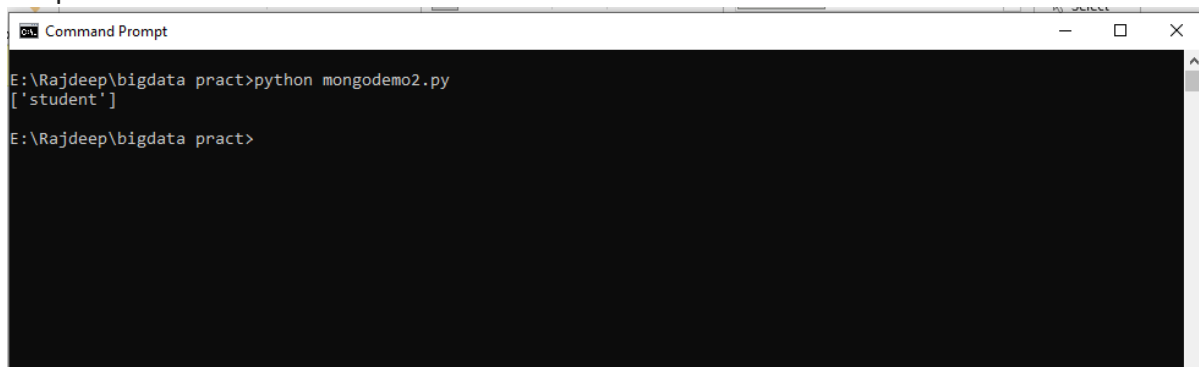


```
Command Prompt
E:\Rajdeep\bigdata pract>python mongodemo1.py
['admin', 'config', 'local', 'mybigdata']
E:\Rajdeep\bigdata pract>
```

Program 2: Creating collection:

```
import pymongo
myclient = pymongo.MongoClient("mongodb://localhost:27017/")
mydb = myclient["mybigdata"]
mycol=mydb["student"]
print(mydb.list_collection_names())
```

Output:

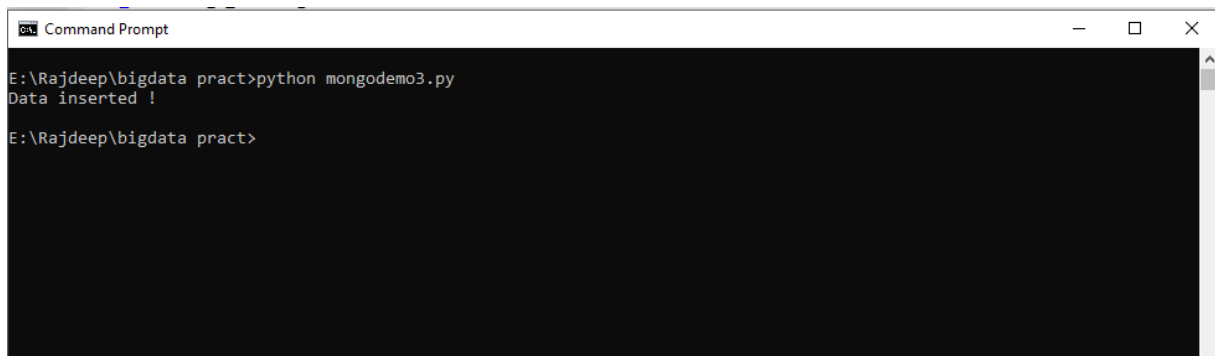


```
Command Prompt
E:\Rajdeep\bigdata pract>python mongodemo2.py
['student']
E:\Rajdeep\bigdata pract>
```

Program 3: Inserting Data

```
import pymongo
myclient = pymongo.MongoClient("mongodb://localhost:27017/")
mydb = myclient["mybigdata"]
mycol=mydb["student"]
mydict={"name":"vai", "address":"bhy"}
x=mycol.insert_one(mydict)
print("Data inserted !")
```

Output:

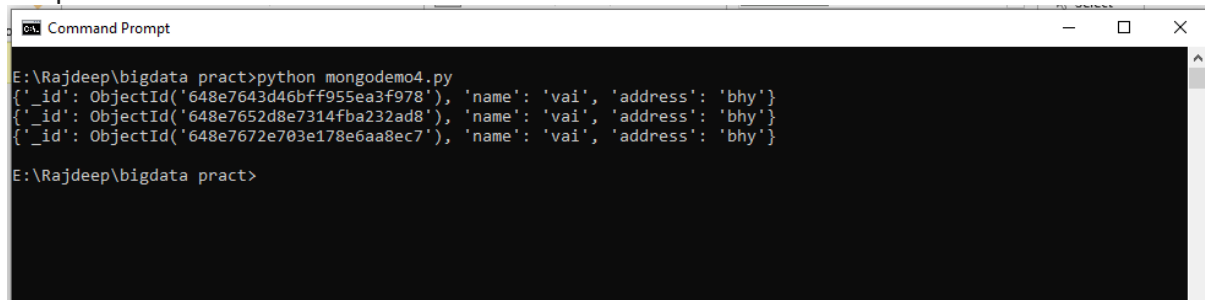


```
Command Prompt
E:\Rajdeep\bigdata pract>python mongodemo3.py
Data inserted !
E:\Rajdeep\bigdata pract>
```

Program 4: Insert Multiple data into Collection

```
import pymongo
myclient = pymongo.MongoClient("mongodb://localhost:27017/")
mydb = myclient["mybigdata"]
mycol=mydb["student"]
mylist=[{"name":"Ganesh", "address":"Mumbai"}, {"name":"Varun", "address":"Mumbai"},
{"name":"Prasoon", "address":"Pune"}, {"name":"Satish", "address":"Pune"},]
x=mycol.insert_many(mylist)
print("Data inserted !")
```

Output:



```
Command Prompt
E:\Rajdeep\bigdata pract>python mongodemo4.py
{'_id': ObjectId('648e7643d46bff955ea3f978'), 'name': 'vai', 'address': 'bhy'}
{'_id': ObjectId('648e7652d8e7314fba232ad8'), 'name': 'vai', 'address': 'bhy'}
{'_id': ObjectId('648e7672e703e178e6aa8ec7'), 'name': 'vai', 'address': 'bhy'}
E:\Rajdeep\bigdata pract>
```

Program 5: Displaying the collection data:

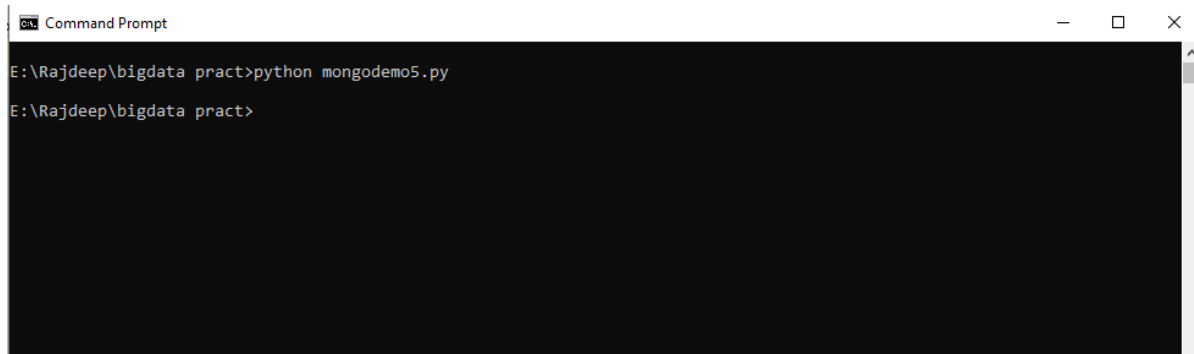
```
import pymongo
myclient = pymongo.MongoClient("mongodb://localhost:27017/")
mydb = myclient["mybigdata"]
mycol = mydb["student"]

myquery = { "name": "Vai" }

mydoc = mycol.find(myquery)

for x in mydoc:
    print(x)
```

Output:



The image shows a screenshot of a Windows Command Prompt window. The title bar at the top reads "Command Prompt" with standard window control buttons (minimize, maximize, close) on the right. The command prompt shows the following text:

```
E:\Rajdeep\bigdata pract>python mongodemo5.py  
E:\Rajdeep\bigdata pract>
```

The background of the command prompt is black, and the text is white. The cursor is positioned at the end of the second line, ready for the next command.

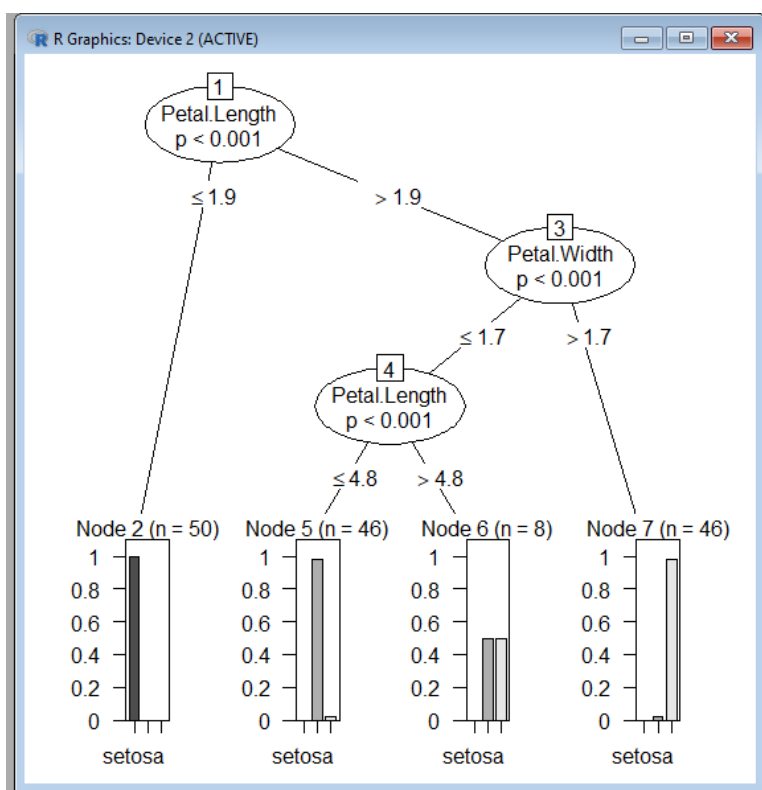
Practical 4

Aim: Implement Decision tree classification techniques

Code-

```
library("party")
print(head(readingSkills))
str(iris)
iris_ctree <- ctree(Species ~ Sepal.Width + Sepal.Length + Petal.Length + Petal.Width,
data=iris)
print (iris_ctree)
plot(iris_ctree)
```

output



Practical 5

Aim: Implement SVM classification

techniques Code-

```
# Importing the dataset

dataset = read.csv(' E:/NIKHILESH/social.csv')


# Selecting relevant columns: Age, EstimatedSalary, Purchased

dataset = dataset[3:5]


# Encoding the target feature as factor

dataset$Purchased = factor(dataset$Purchased, levels = c(0, 1))


# Splitting the dataset into the Training set and Test set

install.packages('caTools') # Run only once

library(caTools)

set.seed(123)

split = sample.split(dataset$Purchased, SplitRatio = 0.75)

training_set = subset(dataset, split == TRUE)

test_set = subset(dataset, split == FALSE)


# Feature Scaling

training_set[-3] = scale(training_set[-3])

test_set[-3] = scale(test_set[-3])


# Fitting SVM to the Training set

install.packages('e1071') # Run only once

library(e1071)

classifier = svm(formula = Purchased ~ .,
                 data = training_set,
                 type = 'C-classification',
```

```
kernel = 'linear')

# Predicting the Test set results
y_pred = predict(classifier, newdata = test_set[-3])

# Making the Confusion Matrix
cm = table(test_set[, 3], y_pred)
print("Confusion Matrix:")
print(cm)

# Visualising the Training set results
install.packages("ElemStatLearn") # Run only once
library(ElemStatLearn)

# Plotting function (for training and test sets)
plot_svm <- function(set, title) {
  X1 = seq(min(set[, 1]) - 1, max(set[, 1]) + 1, by = 0.01)
  X2 = seq(min(set[, 2]) - 1, max(set[, 2]) + 1, by = 0.01)
  grid_set = expand.grid(X1, X2)
  colnames(grid_set) = c('Age', 'EstimatedSalary')
  y_grid = predict(classifier, newdata = grid_set)
  plot(set[, -3],
        main = title,
        xlab = 'Age', ylab = 'Estimated Salary',
        xlim = range(X1), ylim = range(X2))
  contour(X1, X2, matrix(as.numeric(y_grid), length(X1), length(X2)), add = TRUE)
  points(grid_set, pch = '.', col = ifelse(y_grid == 1, 'springgreen3', 'tomato'))
  points(set, pch = 21, bg = ifelse(set[, 3] == 1, 'green4', 'red3'))
}

# Plot training and test results
```



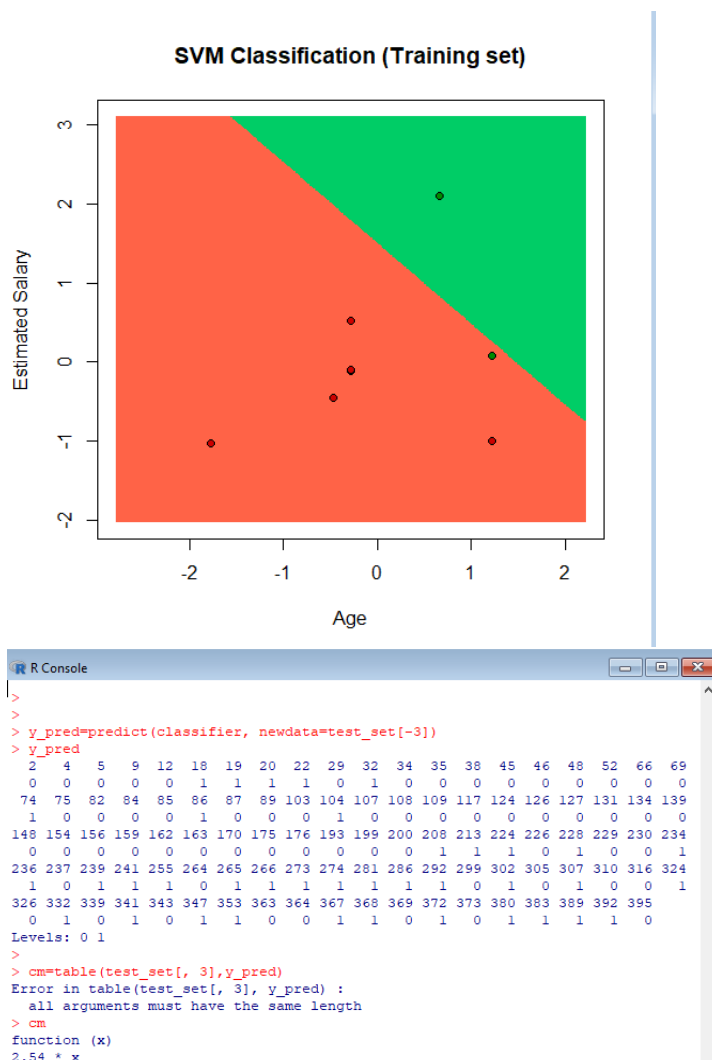
```
plot_svm(training_set, 'SVM Classification (Training set)')
```

```
> set = training_set
> x1 = seq(min(set[, 1]) - 1, max(set[, 1]) + 1, by = 0.01)
> x2 = seq(min(set[, 2]) - 1, max(set[, 2]) + 1, by = 0.01)
```

```
> grid_set = expand.grid(x1, x2)
> colnames(grid_set) = c('Age', 'EstimatedSalary')
> y_grid = predict(classifier, newdata = grid_set)
> plot(set[, -3],
+       main = 'SVM (Training set)',
+       xlab = 'Age', ylab = 'Estimated salary',
+       xlim = range(x1), ylim = range(x2))
```

```
> contour(x1, x2, matrix(as.numeric(y_grid), length(x1), length(x2)), add = TRUE)
> points(grid_set, pch = '.', col = ifelse(y_grid == 1, 'coral1', 'aquamarine'))
> points(set, pch = 21, bg = ifelse(set[, 3] == 1, 'green4', 'red3'))
```

Output:



Practical 6

Aim : REGRESSION MODEL Import a data from web storage. Name the dataset and now do Logistic Regression to find out relation between variables that are affecting the admission of a student in an institute based on his or her GRE score, GPA obtained and rank of the student. Also check the model is fit or not. require (foreign), require(MASS).

Linear regression practical

code-

```
# Load dataset
```

```
college <-
```

```
read.csv("https://raw.githubusercontent.com/ropensci/datapack/main/inst/extdata/pkg-example/binary.csv")
```

```
head(college)
```

```
nrow(college)
```

```
# Install and load caTools
```

```
install.packages("caTools") # Run only once
```

```
library(caTools)
```

```
# Split dataset
```

```
set.seed(123)
```

```
split <- sample.split(college$admit, SplitRatio = 0.75)
```

```
training_reg <- subset(college, split == TRUE)
```

```
test_reg <- subset(college, split == FALSE)
```

```
# Fit logistic regression model
```

```
fit_logistic_model <- glm(admit ~ ., data = training_reg, family = "binomial")
```

```
# View coefficients
```

```
coef(fit_logistic_model)["gre"]
```

```
coef(fit_logistic_model)["gpa"]
```

```
coef(fit_logistic_model)["rank"]
```

```
# Predict probabilities on test data
```

```
predict_reg <- predict(fit_logistic_model, newdata = test_reg, type = "response")
```

```
# Plot Conditional Density
```

```
cdplot(as.factor(admit) ~ gpa, data = college)
```

```
cdplot(as.factor(admit) ~ gre, data = college)
```

```
cdplot(as.factor(admit) ~ rank, data = college)
```

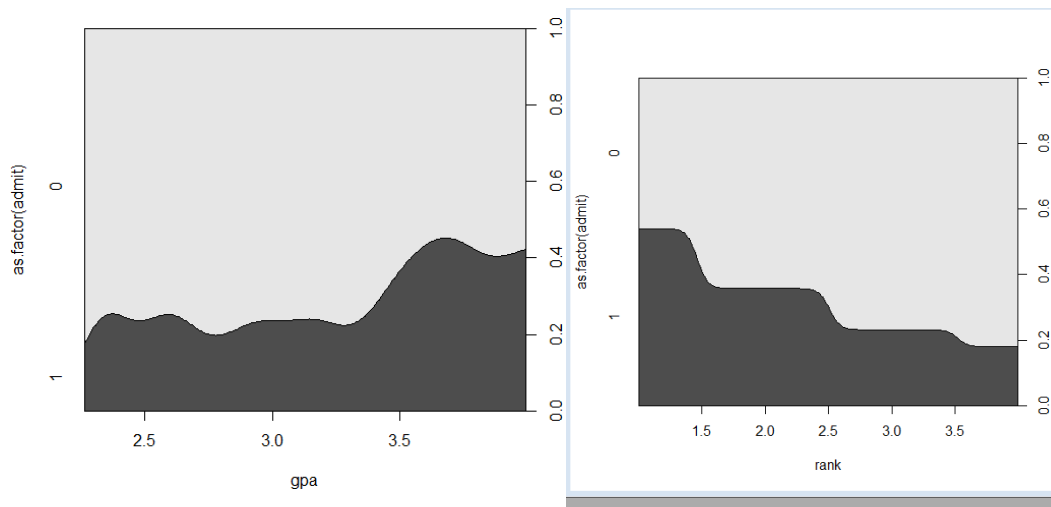
```
# Convert probabilities to binary predictions
```

```
predict_binary <- ifelse(predict_reg > 0.5, 1, 0)
```

Confusion Matrix

```
table(Actual = test_reg$admit, Predicted = predict_binary)
```

output



Practical 7

Aim: MULTIPLE REGRESSION MODEL: Apply multiple regressions, if data have a continuous independent variable. Apply on above dataset.

Code-

Explain Multiple regression in detail.

Load dataset

```
college <- read.csv("https://raw.githubusercontent.com/csquared/udacity-dlnd/master/nn/binary.csv")
```

```
head(college)
```

```
nrow(college)
```

Install and load caTools (only run install once)

```
install.packages("caTools") # Only the first time
```

```
library(caTools)
```

Data splitting

```
set.seed(123)
```

```
split <- sample.split(college$admit, SplitRatio = 0.75)
```

```
training_reg <- subset(college, split == TRUE)
```

```
test_reg <- subset(college, split == FALSE)
```

Fit logistic regression model

```
fit_MRegressor_model <- glm(formula = admit ~ gre + gpa + rank, data = training_reg, family = binomial)
```

Predict probabilities on test set

```
predict_reg <- predict(fit_MRegressor_model, newdata = test_reg, type = "response")
```

```
head(predict_reg)
```

Classify predictions (threshold = 0.5)

```
predict_class <- ifelse(predict_reg > 0.5, 1, 0)
```

Plot conditional density plots

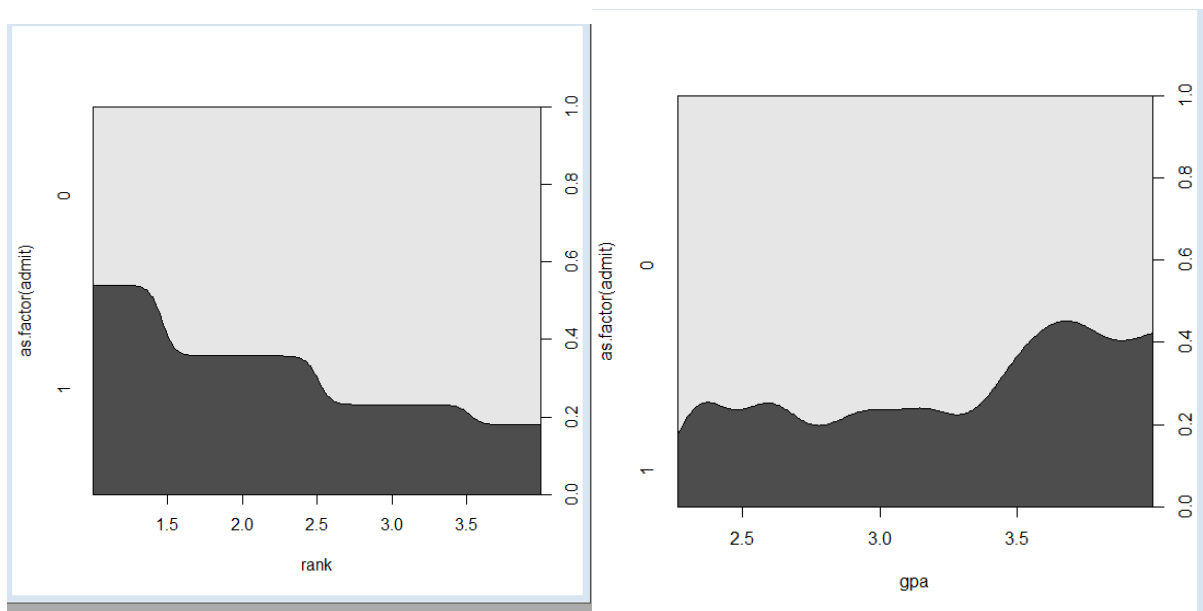
```
cdplot(as.factor(admit) ~ gpa, data = college)
```

```
cdplot(as.factor(admit) ~ gre, data = college)
```

```
cdplot(as.factor(admit) ~ rank, data = college)
```

Confusion matrix

```
table(Actual = test_reg$admit, Predicted = predict_class)
```

Output

Practical 8

Aim : CLASSIFICATION MODEL a. Install relevant package for classification. b. Choose classifier for classification problem. c. Evaluate the performance of classifier.

Navebyse

```
data(iris)
str(iris)
install.packages("e1071")
install.packages("caTools")
install.packages("caret")
library(e1071)
library(caTools)
library(caret)
split <- sample.split(iris, SplitRatio=0.7)
train_c1 <- subset(iris, split=="TRUE")
test_c1 <- subset(iris, split=="FALSE")
train_scale <- scale(train_c1[, 1:4])
test_scale <- scale(test_c1[, 1:4])

set.seed(120)
classifier_c1 <- naiveBayes(Species ~ ., data = train_c1)
classifier_c1

y_pred <- predict(classifier_c1, newdata = test_c1)
cm <- table(test_c1$Species, y_pred)
cm
confusionMatrix(cm) data(iris)
str(iris)
install.packages("e1071")
install.packages("caTools")
install.packages("caret")
library(e1071)
library(caTools)
library(caret)
split <- sample.split(iris, SplitRatio=0.7)
train_c1 <- subset(iris, split=="TRUE")
test_c1 <- subset(iris, split=="FALSE")
train_scale <- scale(train_c1[, 1:4])
test_scale <- scale(test_c1[, 1:4])

set.seed(120)
classifier_c1 <- naiveBayes(Species ~ ., data = train_c1)
classifier_c1
output -
```

```

Naive Bayes Classifier for Discrete Predictors

Call:
naiveBayes.default(x = X, y = Y, laplace = laplace)

A-priori probabilities:
Y
  setosa versicolor virginica
0.3333333 0.3333333 0.3333333

Conditional probabilities:
  Sepal.Length
Y      [,1]      [,2]
setosa  5.050000 0.3739353
versicolor 5.833333 0.5141671
virginica  6.626667 0.5545631

  Sepal.Width
Y      [,1]      [,2]
setosa  3.443333 0.3838852
versicolor 2.753333 0.3191944
virginica  2.976667 0.3058998

```

```
y_pred <- predict(classifier_c1, newdata= test_c1)
```

```
cm <- table(test_c1$Species, y_pred)
```

```
cm
```

```
confusionMatrix(cm)
```

output-

```

cm
      y_pred
      setosa versicolor virginica
setosa      20          0          0
versicolor   0         18          2
virginica    0          1         19
confusionMatrix(cm)

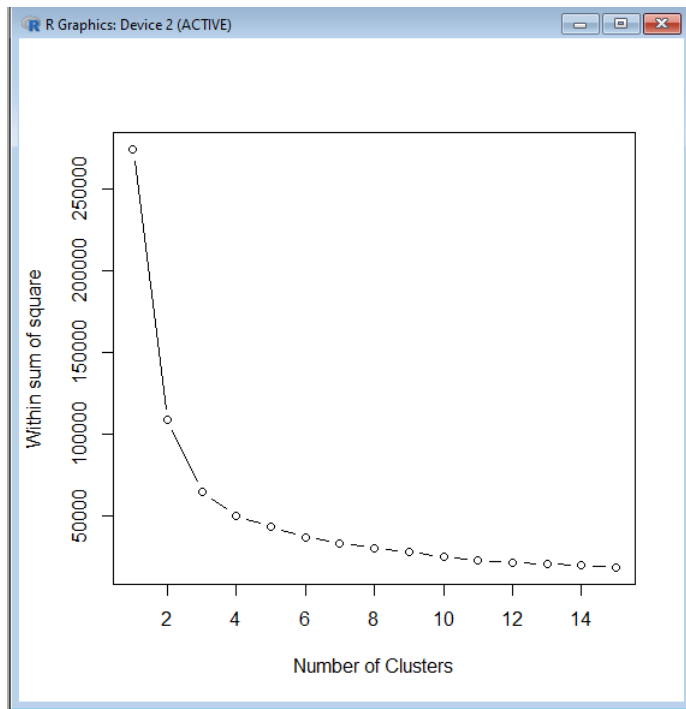
```

Practical 9

Aim : CLUSTERING MODEL a. Clustering algorithms for unsupervised classification.

b. Plot the cluster data using R visualizations.

```
install.packages("plyr")
install.packages("ggplot2")
install.packages("cluster")
install.packages("lattice")
install.packages("grid")
install.packages("gridExtra")
library(plyr)
library(ggplot2)
library(cluster)
library(lattice)
library(grid)
library(gridExtra)
grade_input=as.data.frame(read.csv("E:/Rajdeep/bigdata
pract/dataset/grades_km_input.csv"))
kmdata_orig=as.matrix(grade_input[, c ("Student","English","Math","Science")])
kmdata=kmdata_orig[,2:4]
kmdata[1:10,]
wss=numeric(15)
for(k in 1:15)wss[k]=sum(kmeans(kmdata,centers=k,nstart=25)$withinss)
plot(1:15,wss,type="b",xlab="Number of Clusters",ylab="Within sum of square")
km = kmeans(kmdata,3,nstart=25)
km
c( wss[3] , sum(km$withinss))
df=as.data.frame(kmdata_orig[,2:4])
df$cluster=factor(km$cluster)
centers=as.data.frame(km$centers)
g1=ggplot(data=df, aes(x=English, y=Math, color=cluster )) +
geom_point() + theme(legend.position="right") +
geom_point(data=centers,aes(x=English,y=Math, color=as.factor(c(1,2,3))),size=10,
alpha=.3, show.legend =FALSE)
g2=ggplot(data=df, aes(x=English, y=Science, color=cluster )) +
geom_point () +geom_point(data=centers,aes(x=English,y=Science,
color=as.factor(c(1,2,3))),size=10, alpha=.3, show.legend=FALSE)
g3 = ggplot(data=df, aes(x=Math, y=Science, color=cluster )) +
geom_point () + geom_point(data=centers,aes(x=Math,y=Science,
color=as.factor(c(1,2,3))),size=10, alpha=.3, show.legend=FALSE)
tmp=ggplot_gtable(ggplot_build(g1))
grid.arrange(arrangeGrob(g1 + theme(legend.position="none"),g2 +
theme(legend.position="none"),g3 + theme(legend.position="none"),top ="High School
Student Cluster Analysis" ,ncol=1))
```

Aprori Practical

code-

```
library(arules)
library(arulesViz)
library(RColorBrewer)
```

```
data(Groceries)
Groceries
```

```
summary(Groceries)
class(Groceries)
```

```
rules = apriori(Groceries, parameter = list(supp = 0.02, conf = 0.2))
summary(rules)
```

```
inspect(rules[1:10])
```

```
arules::itemFrequencyPlot(Groceries, topN = 20,
col = brewer.pal(8, 'Pastel2'),
main = 'Relative Item Frequency Plot',
type = "relative",
ylab = "Item Frequency (Relative)")
```

```
itemsets = apriori(Groceries, parameter = list(minlen=2, maxlen=2,support=0.02,
target="frequent itemsets"))
summary(itemsets)
inspect(itemsets)
```

```
itemsets_3 = apriori(Groceries, parameter = list(minlen=3, maxlen=3,support=0.02,  
target="frequent itemsets"))  
summary(itemsets_3)
```

```
inspect(itemsets_3)
```

output-

